

GravityPack: Exploring a Wearable Gravity Display for Immersive Interaction Using Liquid-based System

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ABSTRACT

Previous works have done several kinds of haptic techniques for simulating the touching experience of the virtual object. However, the feedback on the object's weight has less been explored. This paper presents GravityPack, a wearable gravity display to simulate grabbing, holding, and releasing the virtual object in the virtual world using the liquid-based system consisting of pumps, pipes, valves, a water tank, and water packs. This system can provide a wide weight range from 110g to 1.8 kg in 40 seconds. Additionally, we design and investigate the visual feedback of weight transition for the delay time of liquid transfer to understand the feasibility of visualization by a user study. With the revealing of design consideration and implementation, the paper also shows the potential use of the liquid-based system and its possibility of the visualization technique to simulate the weight sensations.

CCS CONCEPTS

• Human-centered computing → Haptic devices; • Hardware → Emerging interfaces.

KEYWORDS

Weight Sensations, Liquid-based Haptic, Virtual Reality

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1 INTRODUCTION

To communicate the perception of an object's weight, previous works have utilized (1) leverage or (2) additional force techniques to simulate weight feedback in virtual reality. Gravity Grabber [Minamizawa et al. 2007] utilizes the deformation on fingerpads to generate a cutaneous, which presents the weight sensation of a virtual object. Shifty [Zenner and Krüger 2017] introduced a weight-shifting dynamic haptic proxy that can automatically change its internal weight distribution to provide haptic feedback. [Fujinawa

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Figure 1: dual water bags attach to (a) single arm, (b) single leg, (c) four water bags attach to limbs, (d) system overview

et al. 2017] present several hand-held VR controllers with weights attached to them in various ways. Thor's Hammer [Heo et al. 2018] presented a force feedback device that can generate a strong thrust of air to simulate weight sensation. These devices effectively provide users with weight feedback and enable them to immerse themselves properly in virtual environment (VE). However, the device itself has a basic weight, and it is difficult to simulate the change from a miniature weight to a mass volume. Additionally, the force feedback from fans or air jets is directional. It may fade away due to wind attenuation, and only the specific parts of the users' body can perceive the force.

Few studies utilize liquid to provide tactile sensations. [Nakano et al. 2006] let user feels a sense of catching goblin through vibration and weight changes. [Niiyama et al. 2014] change the weight of a target object by moving liquid metal to an external reservoir, which combined with a dynamic volume-changing function to simulate the density of different materials. GravityCup [Cheng et al. 2018] is a liquid-based device using the increment and decrement of liquid weight to simulate the weight change. FlowGlove [Liu et al. 2020] is a hand-worn device that provides pressure and vibration sensations with a liquid bladder filled with water. Due to the fluidity and deformability of the liquid, it can be delivered to multiple parts of the body, giving users a wide range of weight sensations. In addition, it can simulate weights from close to empty to heavier

weights by dynamically adjusting the amount of water. Therefore, we utilize water as the material that provides weight to our system.

2 DESIGN AND IMPLEMENTATION

Based on the characteristic of liquids, we have designed 4 fixable water bags that can be worn on the limbs, connected to the piping system with a 120cm long 80g hose. The micro controller will send on/off signals to 2 17A motor drivers and 5 TB6612FNG motor drivers via Bluetooth. The former controls the pump for delivery and drawing water, and the latter controls ten water valves. By turning on the pump and water valve, the water will be delivered from the water tank to the corresponding water bags on the limbs to increase the weight. Alternatively, turn on the drawing pump and valves to draw water from the water bag back to the tank to reduce the weight. In this way, we can dynamically adjust the amount of water to achieve the purpose of weight change. The water delivery rate of a bag from 0 cm height is 46.9 g/s, and the drawn rate is 39.2 g/s; at 180 cm height, the water delivery rate is 39.7 g/s, and the drawing rate is 42.6 g/s.

3 USER STUDY

To explore the possibilities of the weight simulation with GravityPack and its visual effects in VR, 14 participants (6 males) were recruited, whose ages ranged from 20-25 years old (mean=22.7, SD=1.54); 12 participants had worn VR headset before. We asked them to wear a water bag on their arm and a VIVE HMD with the noise-canceling headphone. The VE was designed as a feeding rabbit scenario. Users need to use the controller trigger to pick up a water bottle to fill the water, then water the carrot. When the carrot grows, users need to pull it up and feed it to the rabbit. We design 4 user interfaces (1) BaseLine(BL): no visual effect when the weight transfers. (2) Light Flow(LF): When the weight transfers, a beam of light flows from the weight source to the object, and it will disappear after reaching the specified weight. (3) Fade In/Out(FIAO): objects gradually appear or disappear as the weight transitions. (4) Slow Motion(SM): the physical hand with the object will move slowly in the direction of the virtual hand when the weight transfer. Until the water bag reaches a specified weight, the physical hand will return to normal speed, and the virtual hand will disappear. Users can experience each interface until they are satisfied. Finally, participants took off the HMD and filled in a custom questionnaire, which used a 7-point scale (7 = totally agree, 1 = totally disagree). In the end, we interviewed the participants.

4 DISCUSSION AND FUTURE WORK

The enjoyment of LF is high in figure2. 5 users said they could feel a clear connection between objects, such as the soul had extracted, but four said that the change looks like sci-fi, so engagement and realism are low. The score of FIAO is average, 4 users could anticipate what the change would look like, but 4 others said they were unsure how much transparency would be considered complete. The average score for SM is lower than the others because users rate it differently. 5 users thought the extra hand with the slow movement was strange, and 2 users thought it was interesting because there is a trace connecting the weight change and action. Overall, we found that when the weight change situation was in a real scene,

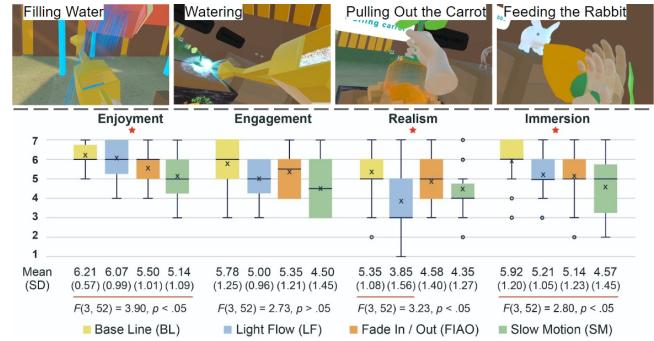


Figure 2: User Study Result

the visual changes of interfaces (LF, FIAO, and SM) appealed to users, which caused users to pay less attention to the weight change. However, the interface can give users an idea of what will happen next. Because BL has no visual effect, it can make people pay more attention to the weight change, and its realism is also the closest to the situation. Therefore, the BL with haptic feedback has a high score. Besides, if we combined BL with haptic and the visual effect, the effect gives users a stronger feeling than the haptic, so the overall score decreases. Consequently, the effect may not be suitable for real-world simulation. In the future, we will make different scenarios, such as weightlessness in the universe and weight change in the magic world. We will explore how the effect of weight change in non-realistic situations can increase the immersive experience.

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